(1) 1.2 -23 ✓

$$\cos\alpha = \frac{v_1}{|v|} \qquad \sin\alpha = \frac{v_2}{|v|}$$

(A) $$\cos\beta = \frac{\omega_1}{|\omega|} \qquad \sin\beta = \frac{\omega_2}{|\omega|}$$

(A) $\theta = \beta - \alpha$

$$\cos\theta = \cos(\beta - \alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha$$

$$= \frac{\omega_1 v_1}{|\omega|\,|v|} + \frac{\omega_2 \omega_2}{|\omega|\,|v|}$$

$$= \frac{\omega_1 v_1 + \omega_2 v_2}{|v| \cdot |\omega|}$$

$$= \frac{\vec{v} \cdot \vec{\omega}}{|\vec{v}| \cdot |\vec{\omega}|}$$

② 1.2 - 28

$u \quad v \quad w$

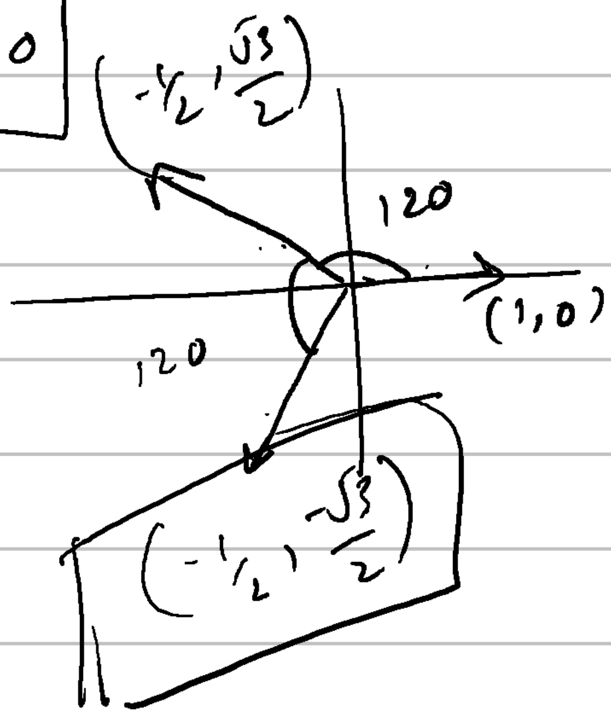
such that

$u \cdot v < 0$ && $u \cdot w < 0$ && $w \cdot v < 0$

$$\left.\begin{aligned} u_1 v_1 + u_2 v_2 &< 0 \\ u_1 w_1 + u_2 w_2 &< 0 \end{aligned}\right\} \checkmark$$

$$v_1 w_1 + v_2 w_2 < 0$$

$\boxed{90 \to 270}$



3 1.3 - 4

$$w_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad w_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \quad w_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

$$x_1 w_1 + x_2 w_2 + x_3 w_3 = 0$$

$$W = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \qquad \boxed{WX = 0}$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 7 & 0 \\ 2 & 5 & 8 & 0 \\ 3 & 6 & 9 & 0 \end{array}\right] \rightarrow \begin{bmatrix} 1 & 4 & 7 \\ 0 & -3 & -6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 4 & 7 \\ 0 & -3 & 6 \\ 0 & -6 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 7 \\ 0 & -3 & 6 \\ 0 & 0 & -17 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 7 \\ 0 & -3 & 6 \\ 0 & 0 & -17 \end{bmatrix}$$

$$-17x_3 = 0$$
$$x_3 = 0$$

→ These vectors are independent.

→ They lie in 3d space.

→ The matrix is invertible.

So wrong!

$$w_1 - 2w_2 + w_3 = \boxed{0}$$

To be solved later with determinants,

(4) 1.3 - 13

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & & 0 \end{bmatrix}$$

$$\begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 0 & 1 & & 1 \end{matrix}$$

1

5 2.1 - 29

$u_0 = (1, 0)$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$

$$A u_0 = u_1 = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$$

$$A u_1 = u_2 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$$

$$= 0.8 \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} + 0.2 \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix}$$

$$= \begin{bmatrix} 0.64 \\ 0.16 \end{bmatrix} + \begin{bmatrix} 0.06 \\ 0.14 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$$

$$= 0.7 \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} + 0.3 \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix}$$

$$= \begin{bmatrix} 0.56 \\ 0.14 \end{bmatrix} + \begin{bmatrix} 0.09 \\ 0.21 \end{bmatrix}$$

$$= \begin{bmatrix} 0.65 \\ 0.35 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 0.65 \\ 0.35 \end{bmatrix}$$

⑥ 2.1 - 20 ✓

$$S \rightarrow \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

$$\begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

$$0.6\begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} + 0.4\begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix}$$

$$= \begin{bmatrix} 0.48 + 0.12 \\ 0.12 + 0.28 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

(7) 2.2 - 20

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & -1 \\ 2 & -1 & 0 \end{bmatrix}$$

$-0 + 4 = 4$ non zero det

If 3^{rd} row is a linear combination of the first two rows then the planes would not have an intersection point

⑧ 2.2 - 30

100 eq^{n} $\qquad$ $\underbrace{A}_{\text{singular}}x = 0$

a) $\underline{0}$

b) $\underbrace{col_1}_{\text{not allowed due to dimension mismatch}} x_1 + col_2 x_2 - - - - - = \underline{\underline{0}}$

$100 \times 100 \quad - \quad 100 \times 1$

$100 \times 1 \quad - \quad 100 \times 1$

c) $(1, 2 \quad \ldots \quad 100)$ $2x -$ $3x - 4x$

9 2.3 - 22

a) $a_{31} x_1 + a_{32} x_2 - - - - a_{3n} x_n$

b) $a_{21} x_1 - a_{11} x_1 \rightarrow x_1 (a_{21} - a_{11})$

c) $a_{21} x_1 - 2 \cdot a_{11} x_1 \rightarrow x_1 (a_{21} - 2 \cdot a_{11})$

d) unchanged

10 2.3 -29

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} E_1 \qquad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} E_2 \qquad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & -1 & -1 & 1 \end{bmatrix} E_3 \qquad \begin{bmatrix} \text{Given} \\ \ldots \end{bmatrix}$$

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

(11) 2.4 - 32

$$\underbrace{A x_1}_{3\times 1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad A x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad A x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} - & - & - \\ - & A - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} | & | & | \\ x_1 & x_2 & x_3 \\ | & | & | \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underline{I}$$

$$\underline{AX = I}$$

(12) 2.4 - 36

$A \to m \times n$

$B \to n \times p$

$C \to p \times q$

$(AB)C \to m \times n \times p + m p q$

$\underbrace{(AB)}_{(m \times p)}$ $(p \times q)$

$= mnp + mpq$

$= mp[n+q]$

$A(BC) = mnq + npq$

$m \times n$ $(n \times q)$

$nq[m+p]$

a) i) $14(4+10) = 14^2 = 196$ ✓✓

ii) $40(2+7) = 360$

b) $(((1 \times n)(n \times 1))$ $p \times n$ ✓ first ow

$((u^T v) w^T)$ ✓

c) $f((AB)C) = mnp + mpq$

$f(A(BC)) = mnq + npq$

$\Rightarrow$ $\underline{f((AB)C) < f(A(BC))}$

✓

13) 2.8 - 7

$$\begin{bmatrix} \text{— row 1 —} \\ \text{— row 2 —} \\ \text{— row1 + row2 —} \end{bmatrix}$$

$A x = 0$

$$\begin{bmatrix} \text{row 1} \\ \text{row 2} \\ \text{row 3} \end{bmatrix} \quad [$$

$A_1 \quad A_2 \quad A_3 \rightarrow$ row vectors

$\Rightarrow$ a)

$$A_1 x = 1$$
$$A_2 x = 0$$
$$A_3 x = 0$$

$$A_3 = A_1 + A_2$$
$$A_1 x + A_2 x = 0$$
$$\boxed{1 + 0 = 0}$$